Assignment #4: Your Choice

Goal

In this assignment, you will deepen your understanding of network protocols by picking one of the C++ projects and completing it. You only need to complete one (completing multiple will not give you extra credit).

DigiPing

Ping is an application that tests whether a particular host is reachable across an IP network. It provides some latency information as well. Our Ping will work by sending an ICMP Echo packet to the destination and then listen for an ICMP Echo Reply (aka “Pong”).

Guessing Game (Client & Server)

A simple guessing game using server and dumb client. The server is responsible for managing all the game state and taking the guesses from the client, responding to the guesser “high” or “low”. The client is responsible for printing output from the server and handling input from the user.

Chat Client

Implement a simple chat client protocol which works with a chat server implementation (provided). The client needs to authenticate, send timely responses to pings, and be able to handle multiple threads of operation. A starter client is provided for you.

Which one should I do?

Ping is simple but requires more low-level socket work. The Guessing Game requires multiple-connection handling and game state management, including input and output. The chat client protocol is straightforward but requires you to work within a UI framework.

Submission

You will submit one ZIP file that includes your solution. Do not include binaries or other directories. Do include a README that informs the grader which project you attempted, and any comments or discoveries of note you ran across.

Grading

See the individual project documentation for further grading and submission instructions.